

# Machine Learning

## CMPSCI 589

**Bruno C. da Silva**

[bsilva@cs.umass.edu](mailto:bsilva@cs.umass.edu)

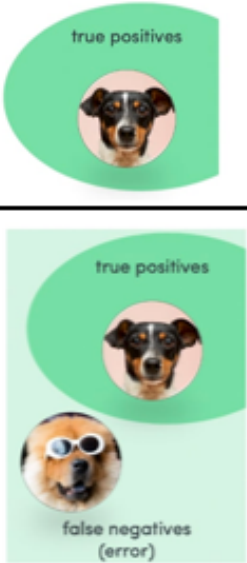
## Ensemble Methods (1/2)

### Bagging Random Forests



# Review: Precision vs Recall

**Precision** =  =  $\frac{TP}{TP + FP}$

**Recall** =  =  $\frac{TP}{TP + FN}$

“Whenever we say it’s a dog,  
we’re usually right”



**Avoiding making wrong detections**

“We’re correctly identifying  
pretty much all dogs that exist”



**Detecting the relevant items/instances**

# Review: Precision vs Recall

## Precision

“Whenever we say it’s a dog,  
we’re usually right”



Avoiding making wrong detections

## Recall

“We’re correctly identifying  
pretty much all dogs that exist”



Detecting the relevant items/instances

- Usually there’s a trade-off between Precision and Recall

Quantifying and optimizing this trade-off

→ F-Score

# Review: F1-Score

## Quantifying and optimizing this trade-off

- **F-Score**: weighted harmonic average of precision and recall
  - where we consider that recall is  $\beta$  times more important than precision

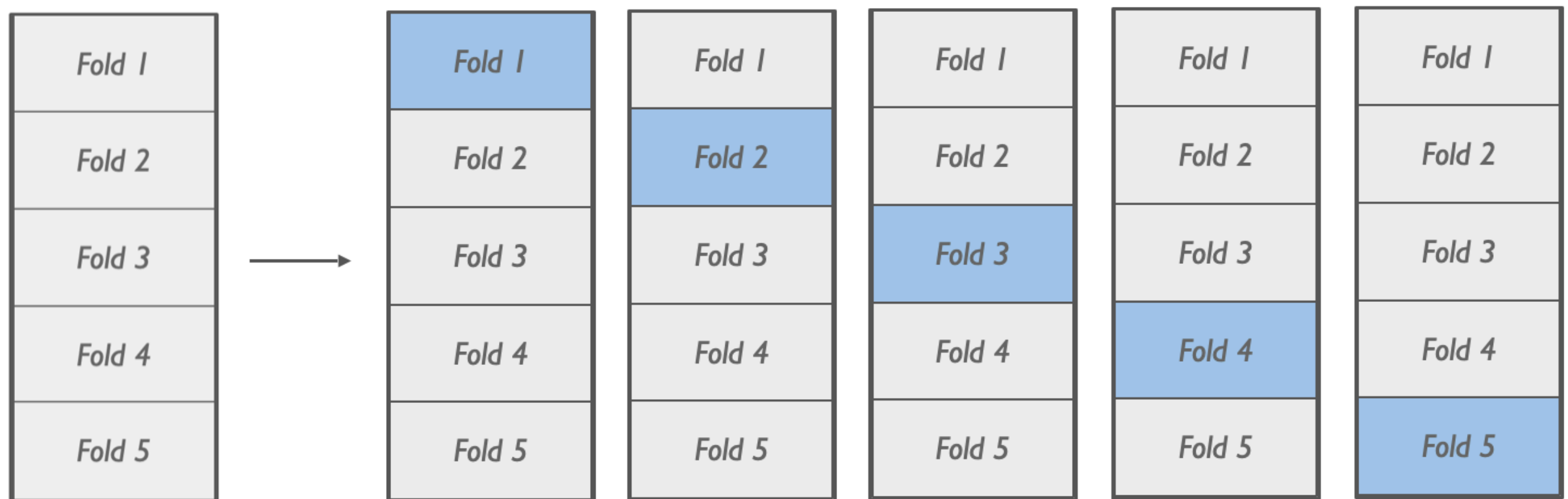
When  $\beta = 1$

→ We give the **same importance** to both precision and recall

$$F_1(f) = 2 \frac{\text{precision}(f) \text{ recall}(f)}{\text{precision}(f) + \text{recall}(f)}$$

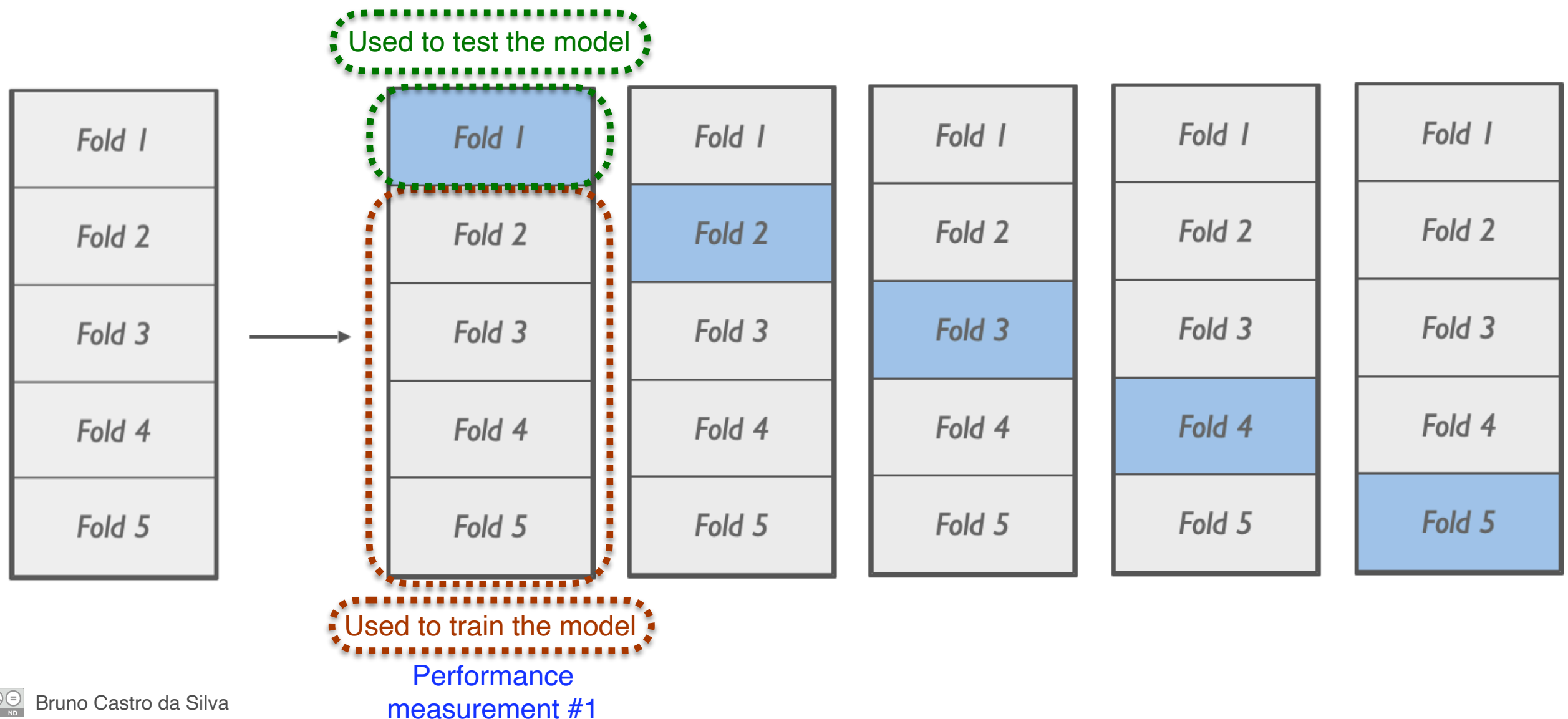
# Review: k-fold Cross-Validation

- Picks  $k-1$  folds and use them to train the model
  - The remaining fold is used as the test set
- Repeat this process  $k$  times
  - Each time, a different fold is used as the test set
- **Model performance** → average performance over the  $k$  folds used to test the model



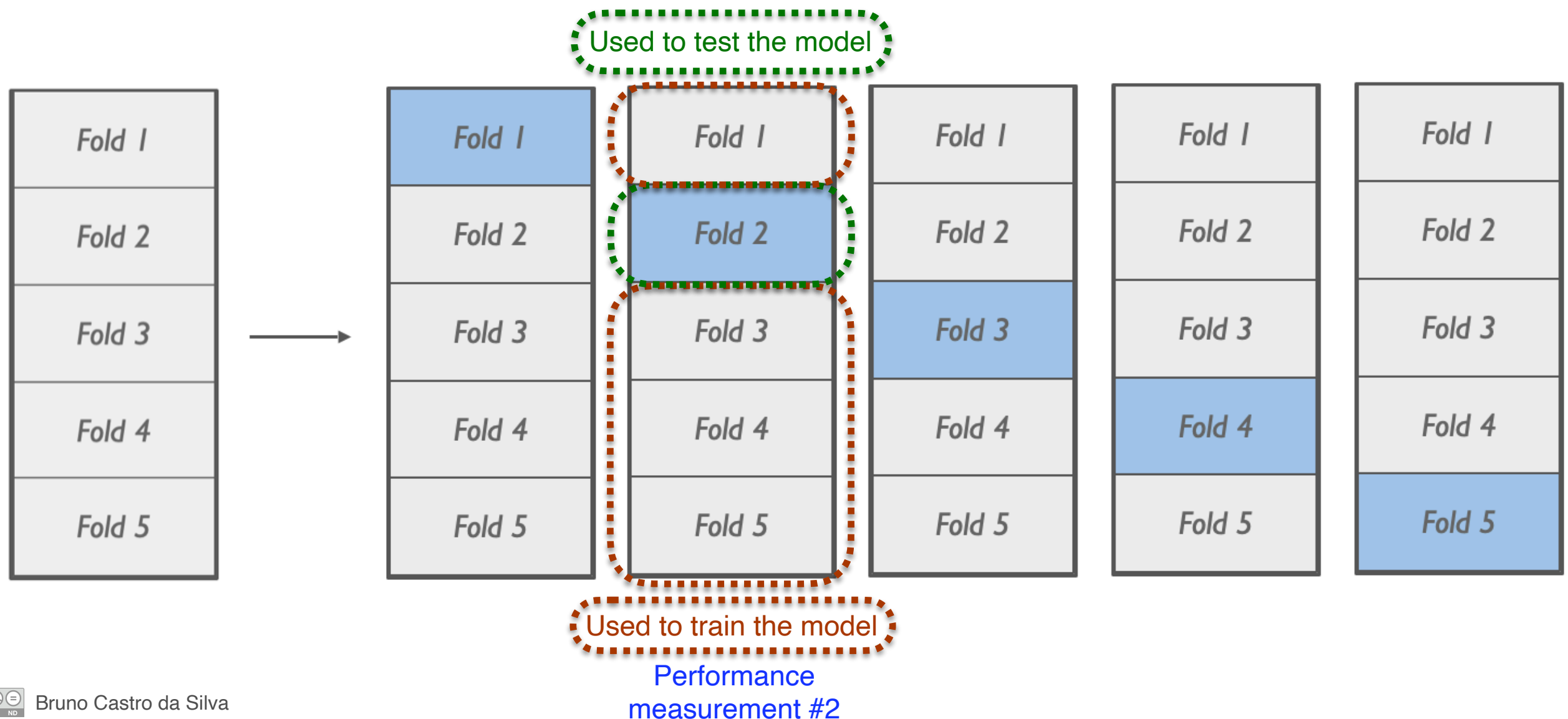
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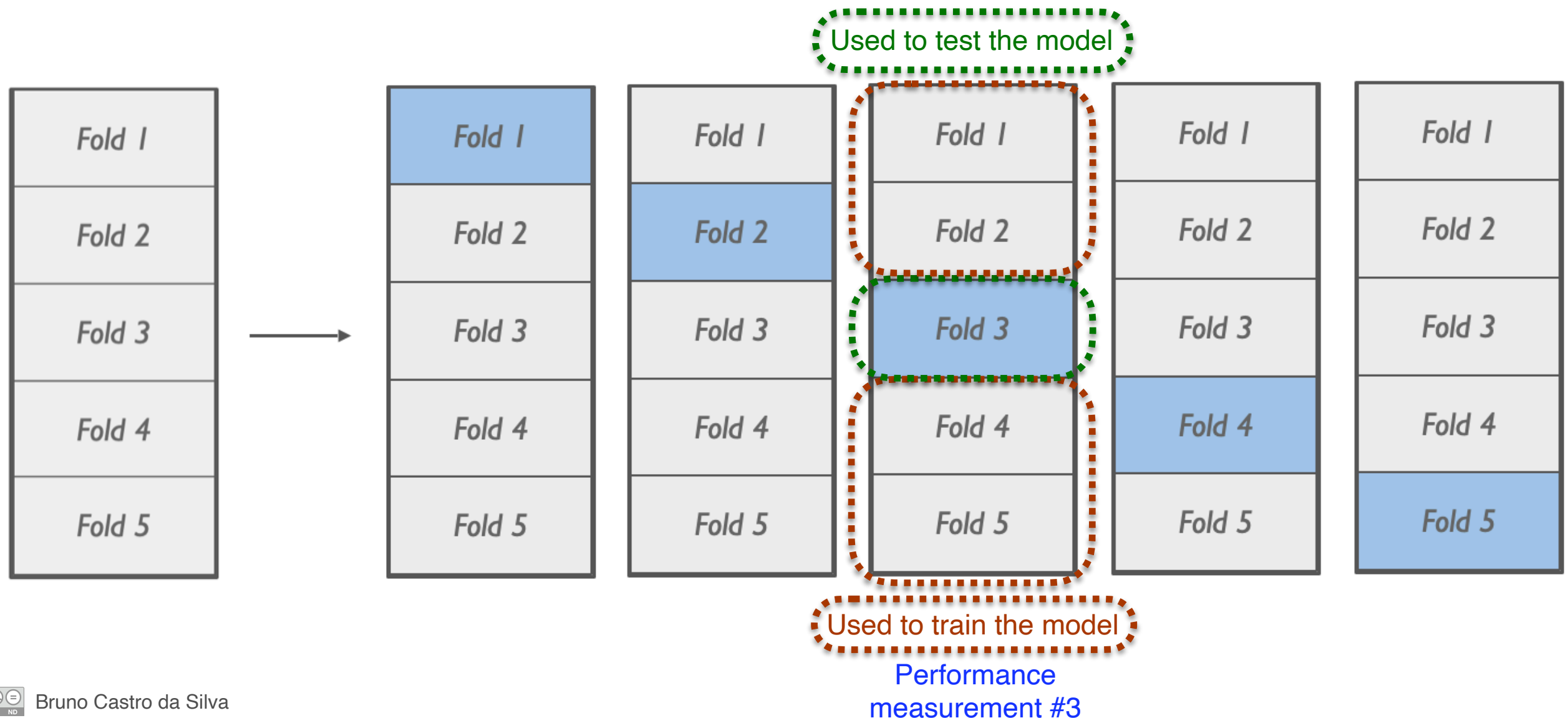
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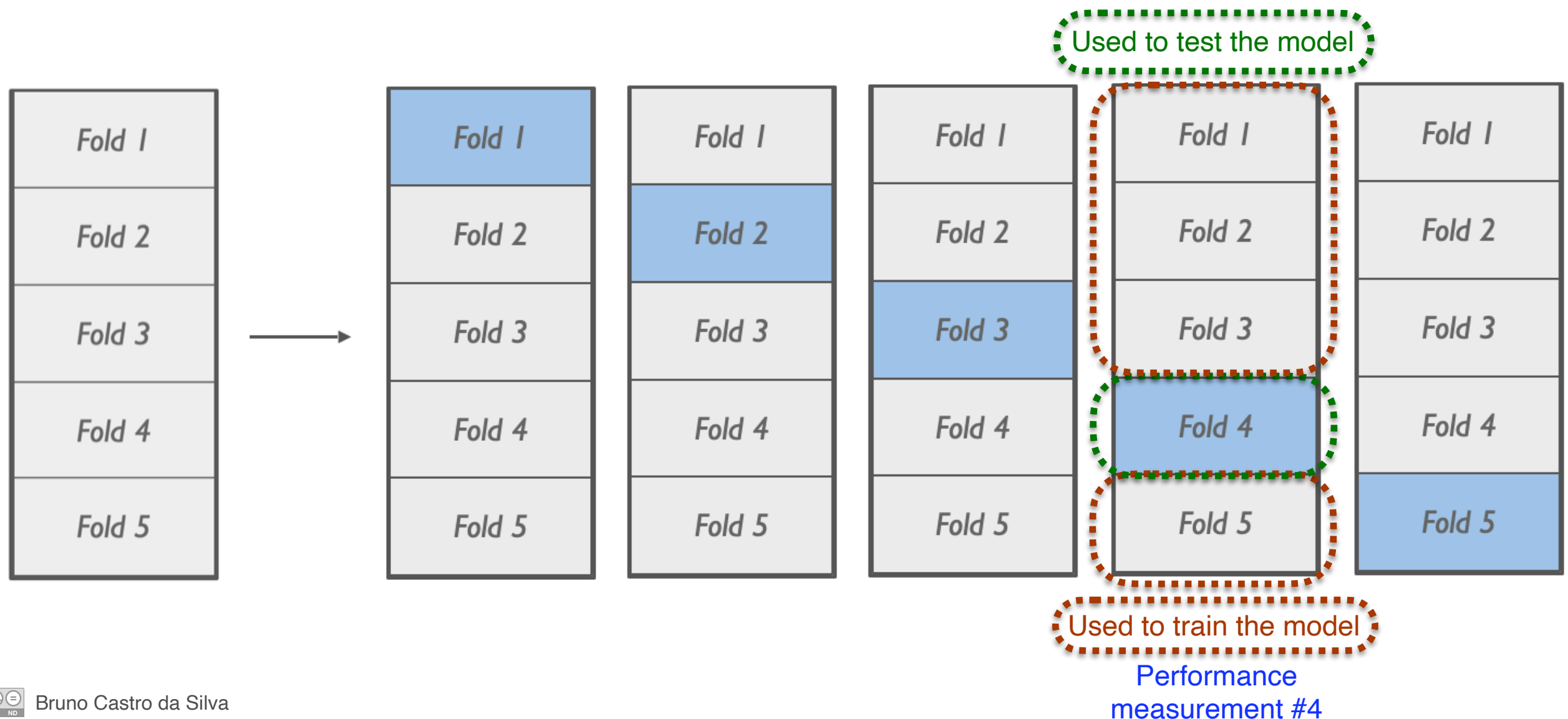
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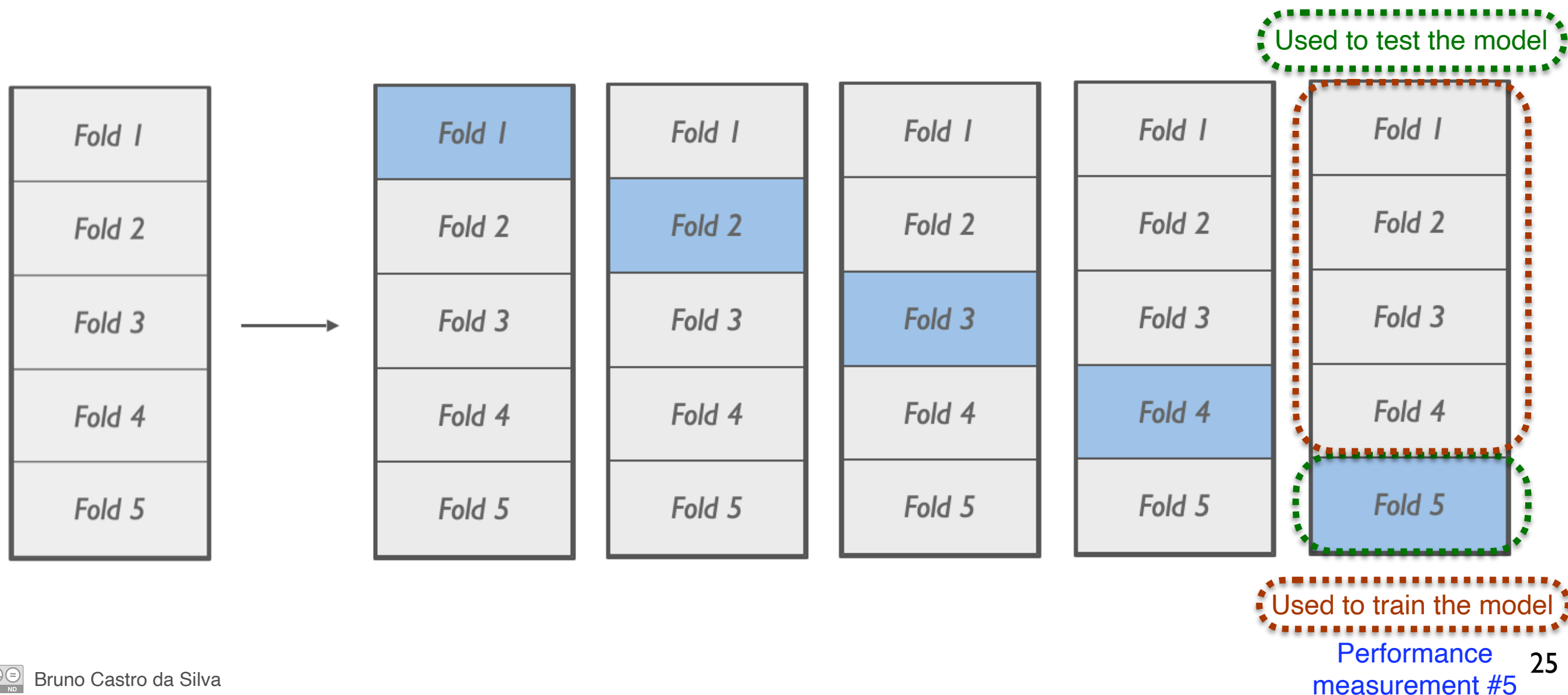
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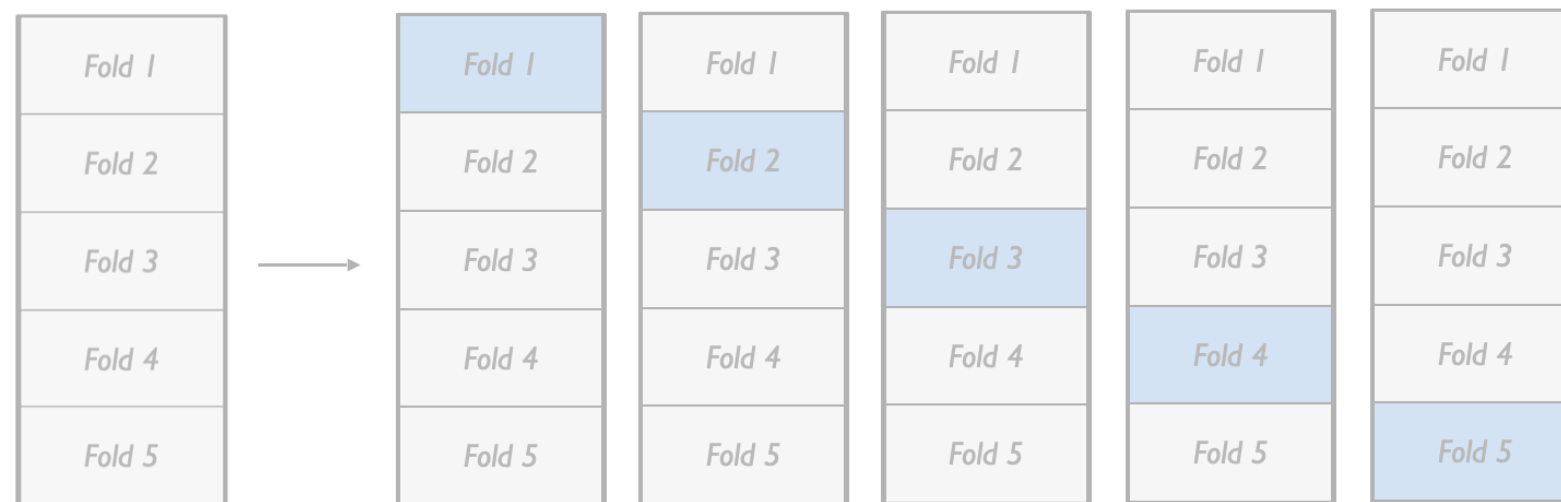
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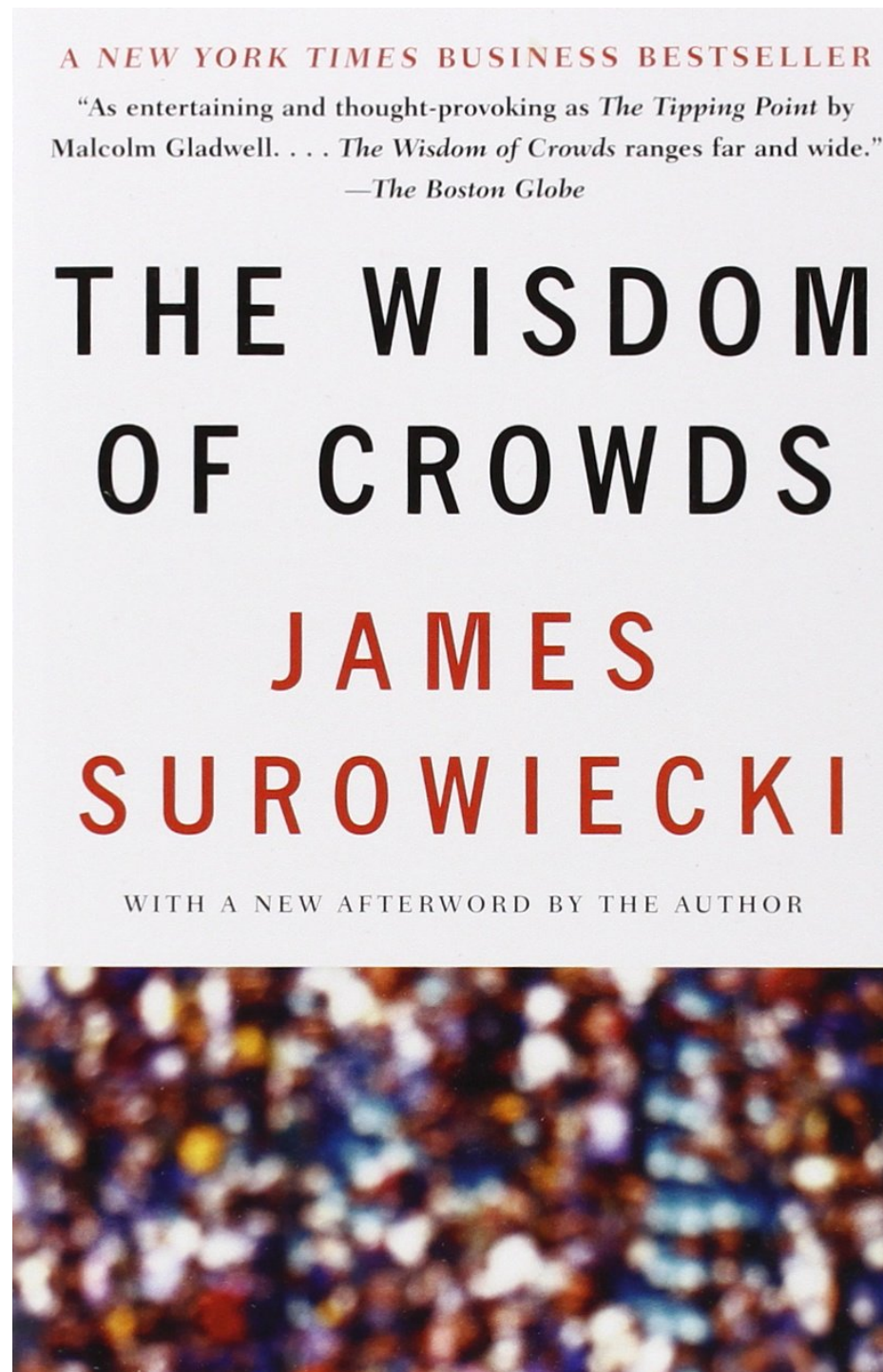


$$\text{Model performance} = (\text{Perf1} + \text{Perf2} + \text{Perf3} + \text{Perf4} + \text{Perf5}) / 5$$

**Typically, k=5 or k=10**

# The Wisdom of Crowds

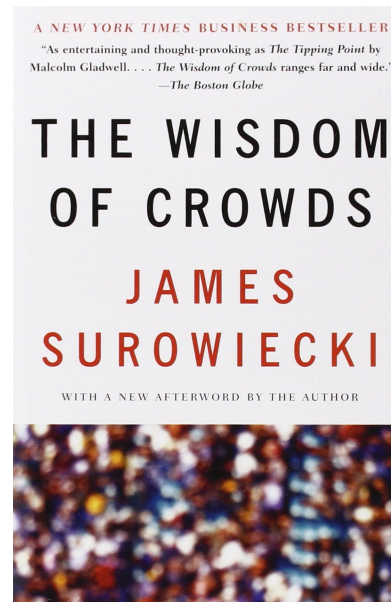
# The Wisdom of Crowds



- Knowledge that emerges from a collective decision
- Often better/“more accurate” than that provided by any one individual person
  - Even (individual) experts!



# The Wisdom of Crowds



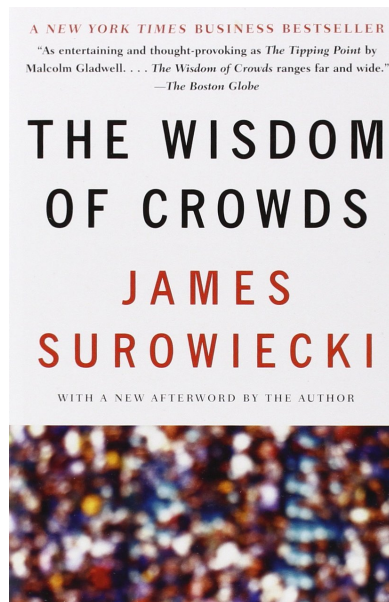
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## Four conditions:

1. Diversity of opinion  
Each person should have private information
2. Independence  
People's opinions are not always determined by the opinions of those around them
3. Decentralization  
No one at the top dictates crowd's answer. People specialize and draw on local knowledge
4. Aggregation  
Some mechanism exists for turning private judgements into a collective decision



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- **Intuitively, in machine learning:**
  - **Many models/algorithms**
  - **Trained independently based on different information/data**

# The Wisdom of Crowds

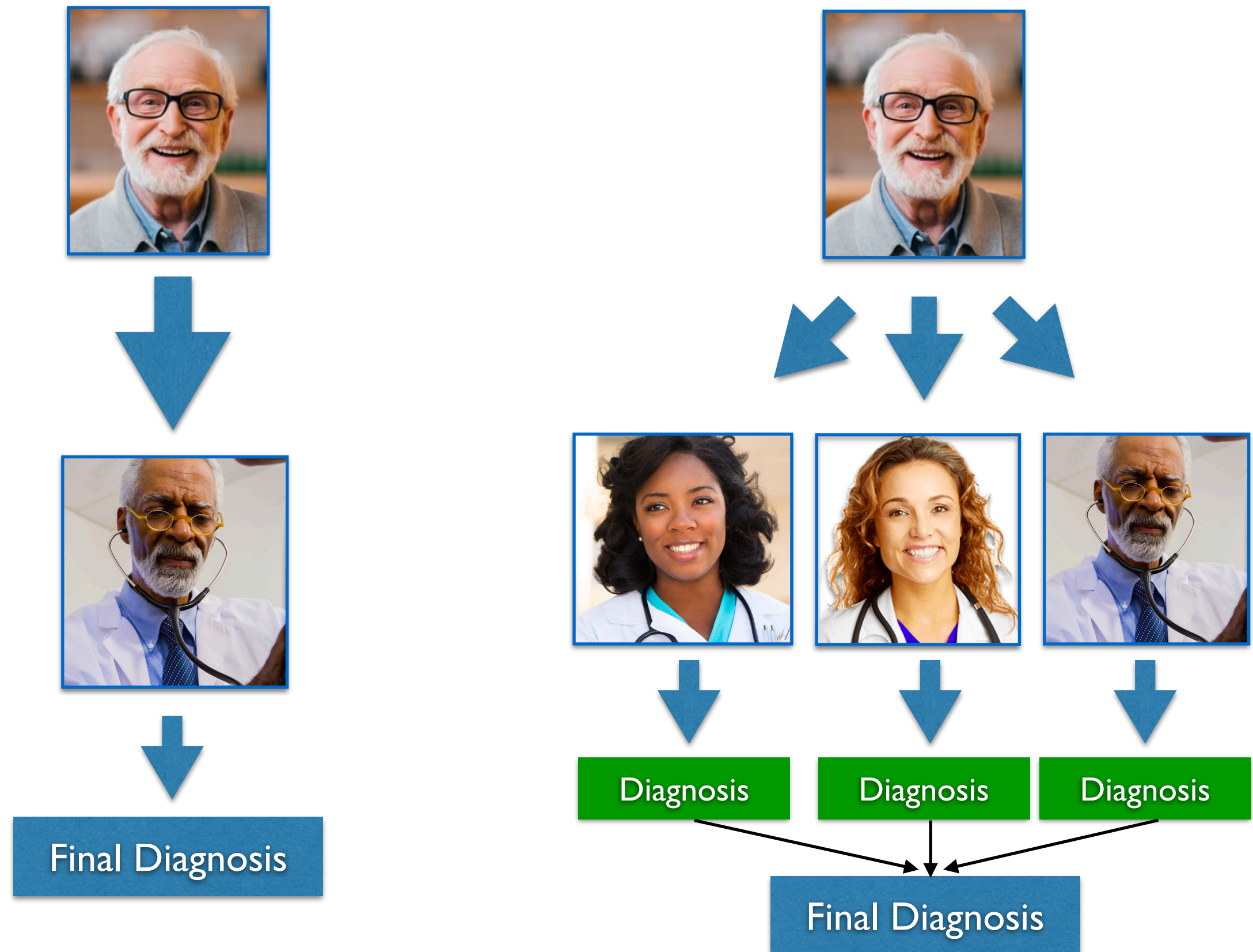
- The **Jelly Beans in a Jar** experiment
  - Michael Mauboussin — 73 Columbia Business School students
  - *“Just by looking at the jar, can you guess how many jelly beans are there?”*
  - The **jar contained 1116 jelly beans**
  - Students would **guess** anywhere from **250 to 4100**
    - **Average error** made by each **student**: 700 (62%)

Average of all 73 guesses: **1151 (3% error)**





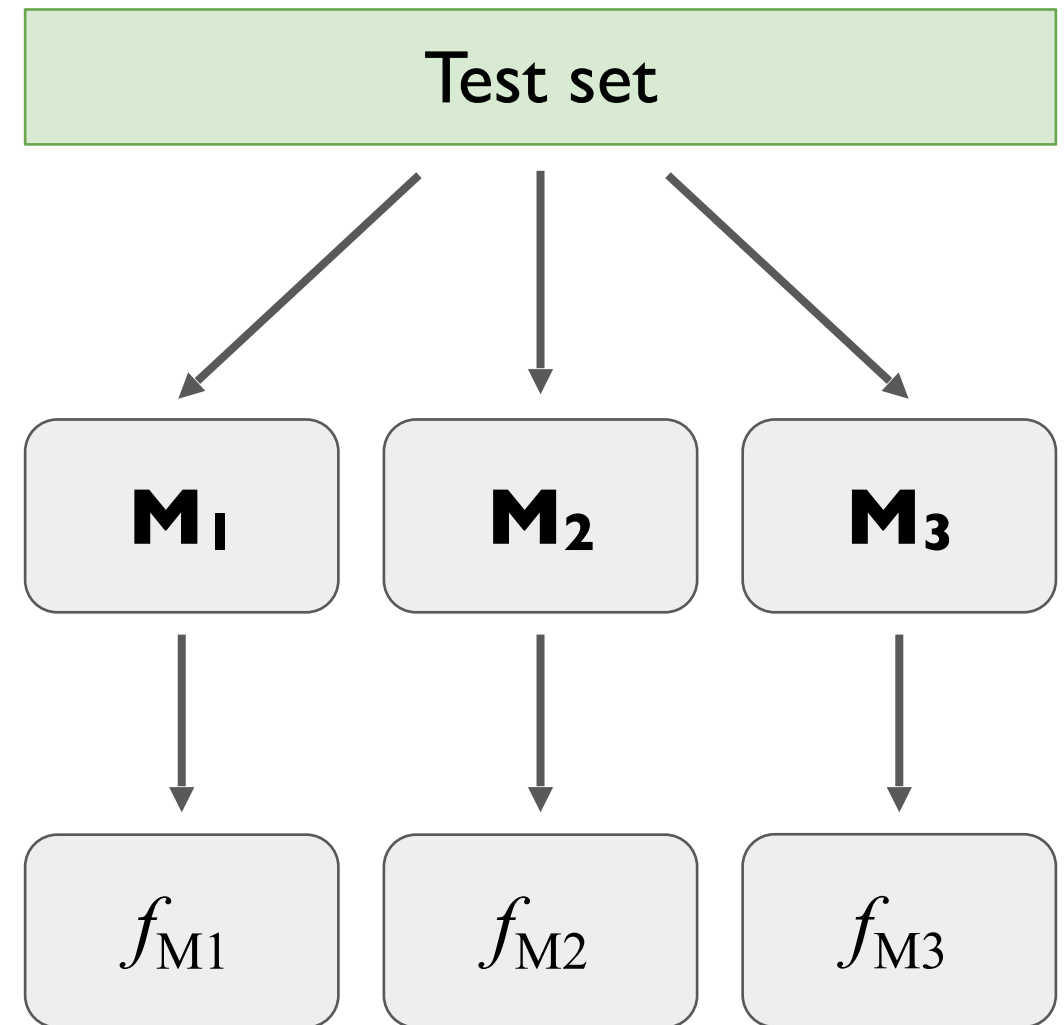
# The Wisdom of Crowds



# Multiple Models in ML

Training set

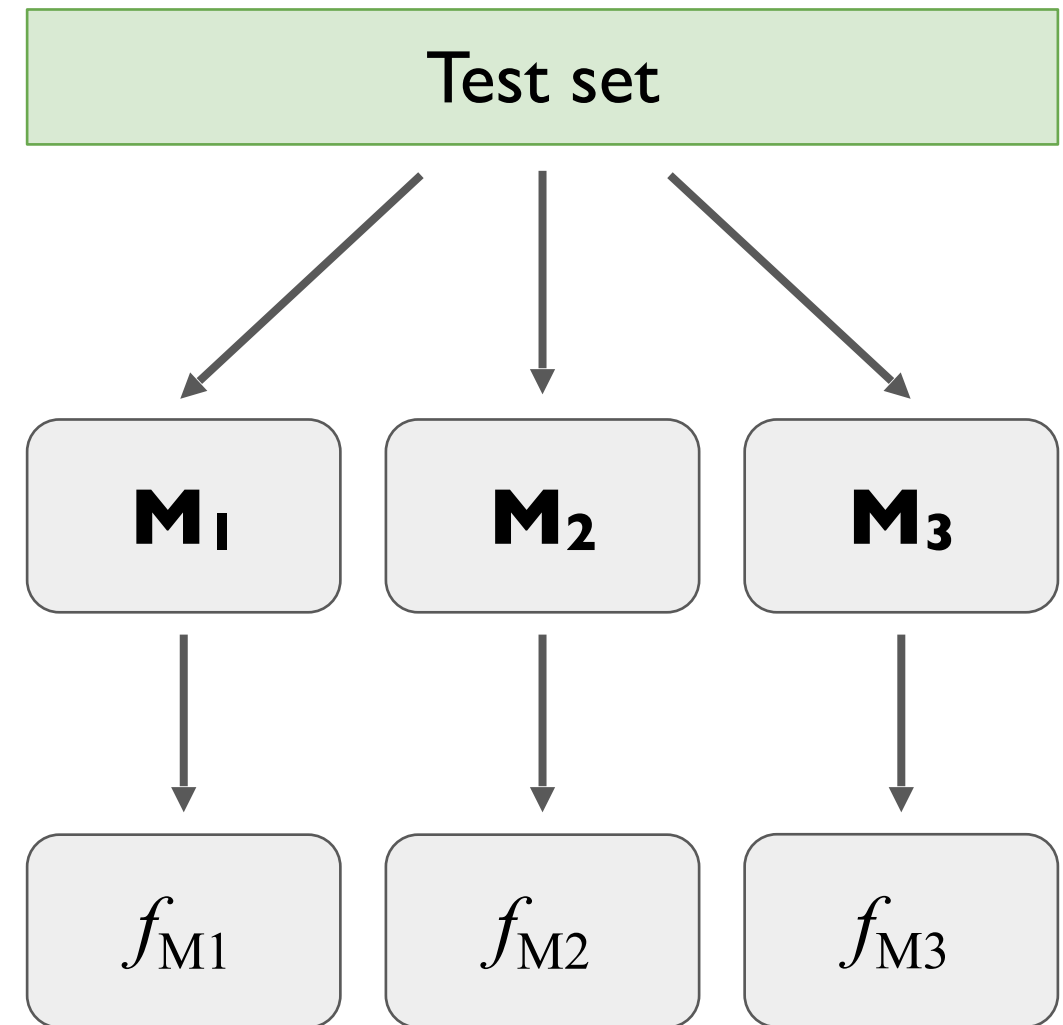
| $\mathbf{x}_k$    | $y_k$ | $f_{M1}$ | $f_{M2}$ | $f_{M3}$ |
|-------------------|-------|----------|----------|----------|
| $\mathbf{x}_1$    | 1     | 1        | 1        | 1        |
| $\mathbf{x}_2$    | 1     | 1        | 0        | 1        |
| $\mathbf{x}_3$    | 1     | 0        | 1        | 1        |
| $\mathbf{x}_4$    | 1     | 1        | 1        | 1        |
| $\mathbf{x}_5$    | 1     | 1        | 1        | 1        |
| $\mathbf{x}_6$    | 1     | 1        | 1        | 0        |
| $\mathbf{x}_7$    | 1     | 0        | 0        | 0        |
| $\mathbf{x}_8$    | 1     | 1        | 1        | 0        |
| $\mathbf{x}_9$    | 1     | 1        | 0        | 1        |
| $\mathbf{x}_{10}$ | 1     | 0        | 1        | 1        |
| Accuracy          |       | 70%      | 70%      | 70%      |



# Multiple Models in ML

Predictions made by different models  
( $M_1, M_2, M_3$ )

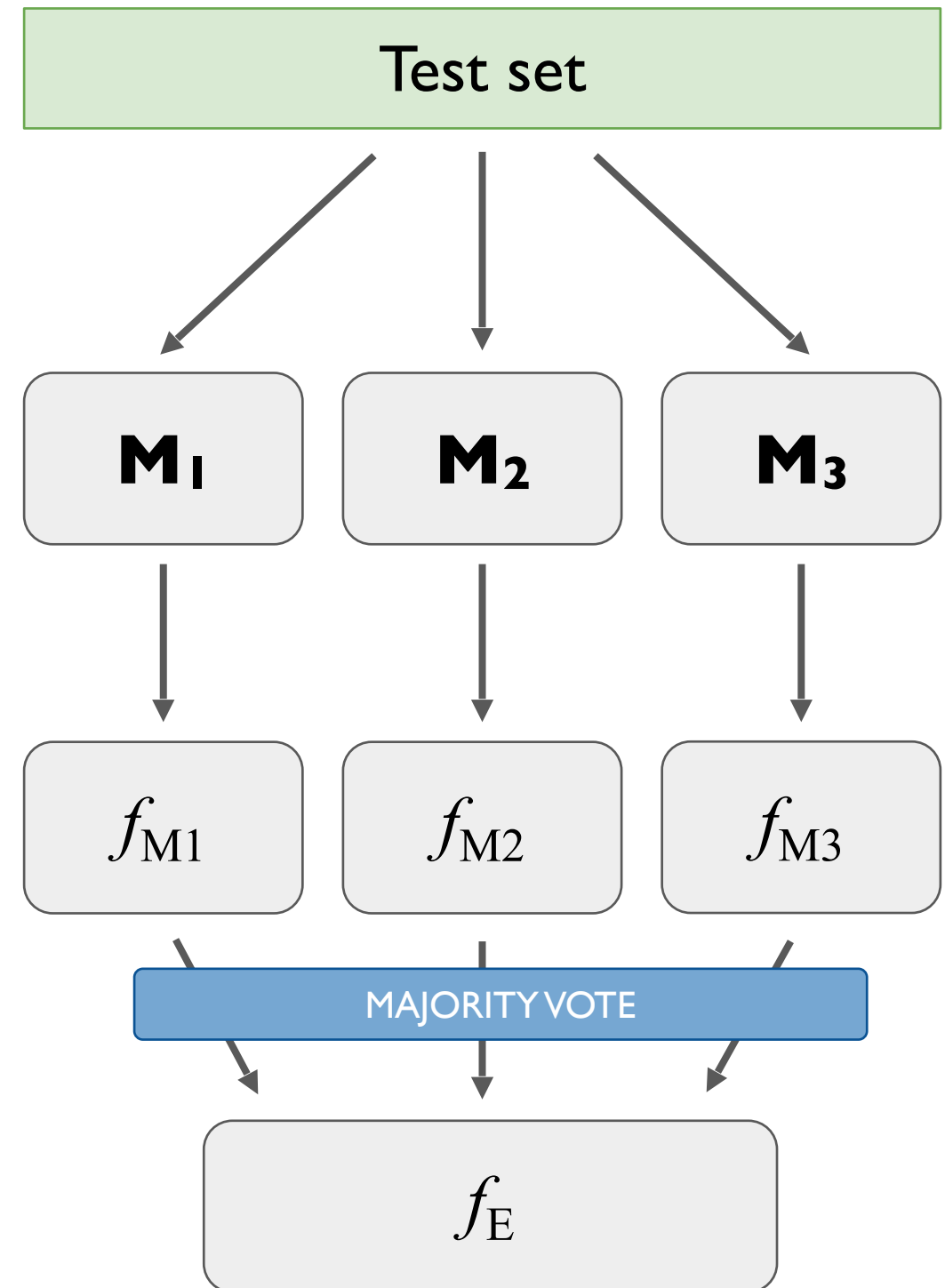
| $\mathbf{x}_k$    | $\mathbf{y}_k$ | $f_{M1}$ | $f_{M2}$ | $f_{M3}$ |
|-------------------|----------------|----------|----------|----------|
| $\mathbf{x}_1$    | 1              | 1        | 1        | 1        |
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| $\mathbf{x}_{10}$ | 1              | 0        | 1        | 1        |
| Accuracy          |                | 70%      | 70%      | 70%      |



# Multiple Models in ML

Combined model (via majority voting)

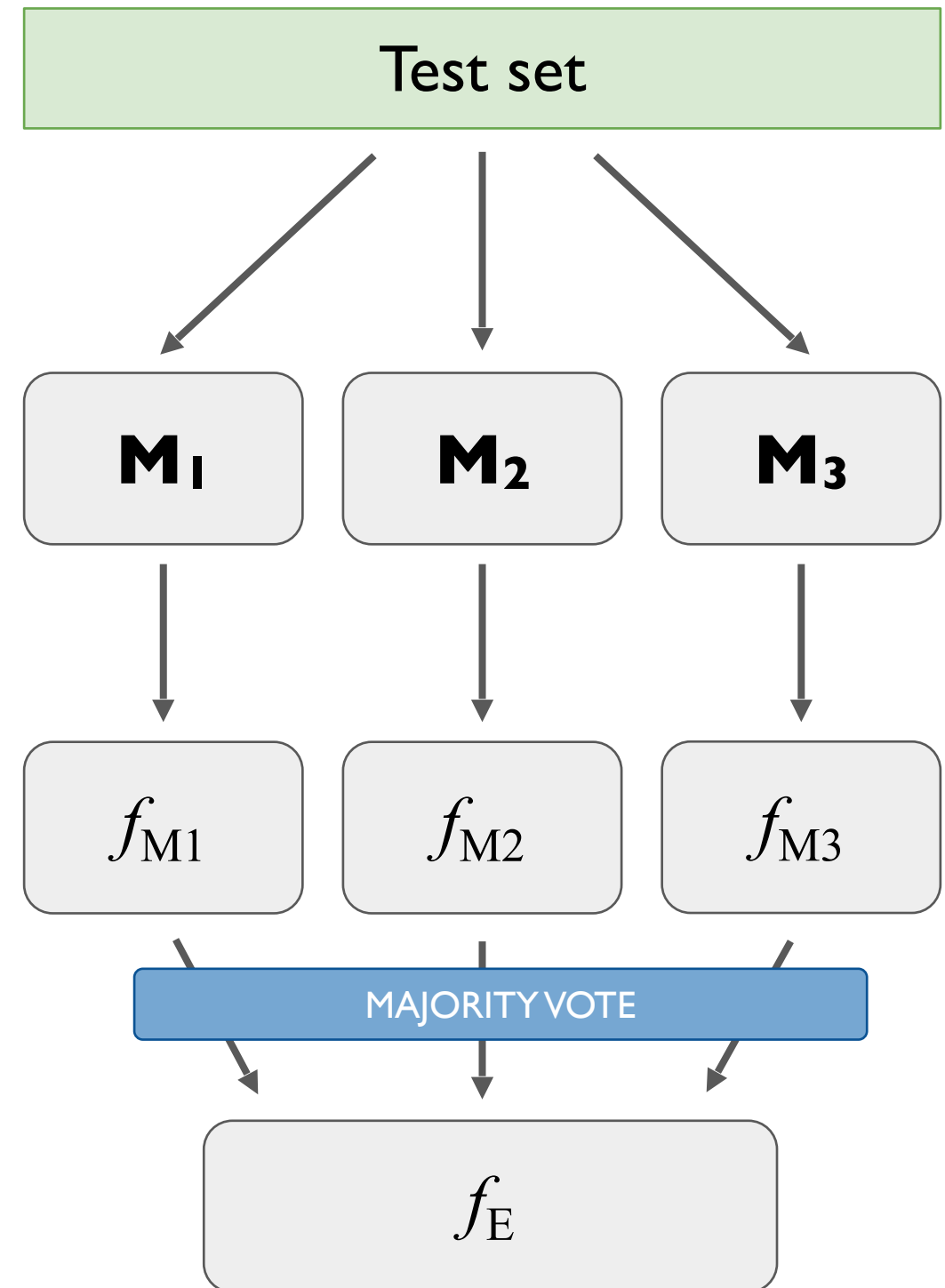
| $\mathbf{x}_k$    | $\mathbf{y}_k$ | $f_{M1}$ | $f_{M2}$ | $f_{M3}$ | $f_E$ |
|-------------------|----------------|----------|----------|----------|-------|
| $\mathbf{x}_1$    | 1              | 1        | 1        | 1        | 1     |
| $\mathbf{x}_2$    | 1              | 1        | 0        | 1        | 1     |
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| $\mathbf{x}_{10}$ | 1              | 0        | 1        | 1        | 1     |
| Accuracy          |                | 70%      | 70%      | 70%      | 90%   |



# Multiple Models in ML

Combined model (via majority voting)

| $\mathbf{x}_k$                                                                                                                         | $\mathbf{y}_k$ | $f_{M1}$ | $f_{M2}$ | $f_{M3}$ | $f_E$ |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------|----------|----------|----------|-------|
| $\mathbf{x}_1$                                                                                                                         | 1              | 1        | 1        | 1        | 1     |
| $\mathbf{x}_2$                                                                                                                         | 1              | 1        | 0        | 1        | 1     |
| $\mathbf{x}_3$                                                                                                                         | 1              | 0        | 1        | 1        | 1     |
| <p>The “consensus” resulting from combining each model’s predictions tends to have higher accuracy than each individual classifier</p> |                |          |          |          |       |
| $\mathbf{x}_8$                                                                                                                         | 1              | 1        | 1        | 0        | 1     |
| $\mathbf{x}_9$                                                                                                                         | 1              | 1        | 0        | 1        | 1     |
| $\mathbf{x}_{10}$                                                                                                                      | 1              | 0        | 1        | 1        | 1     |
| Accuracy                                                                                                                               |                | 70%      | 70%      | 70%      | 90%   |

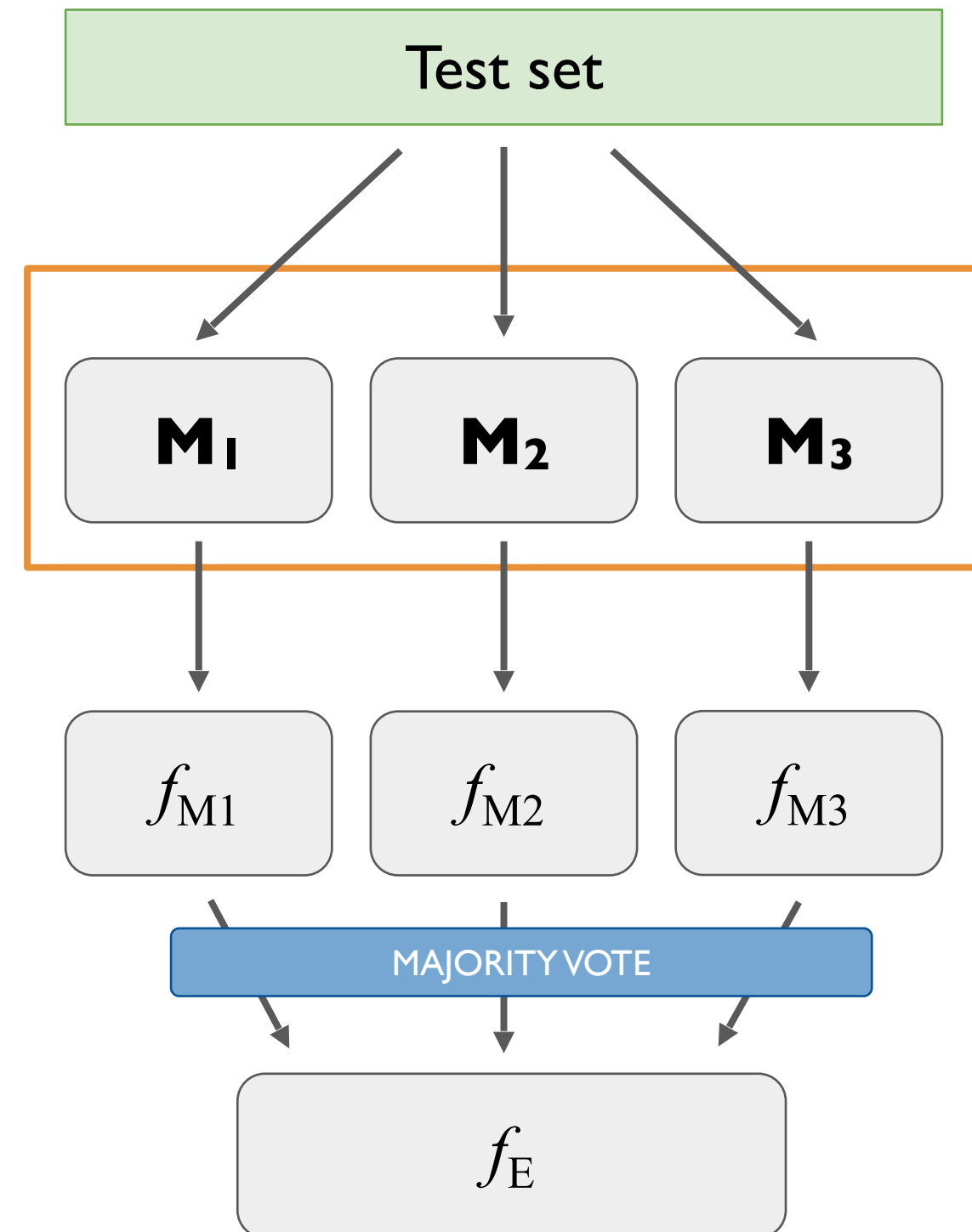


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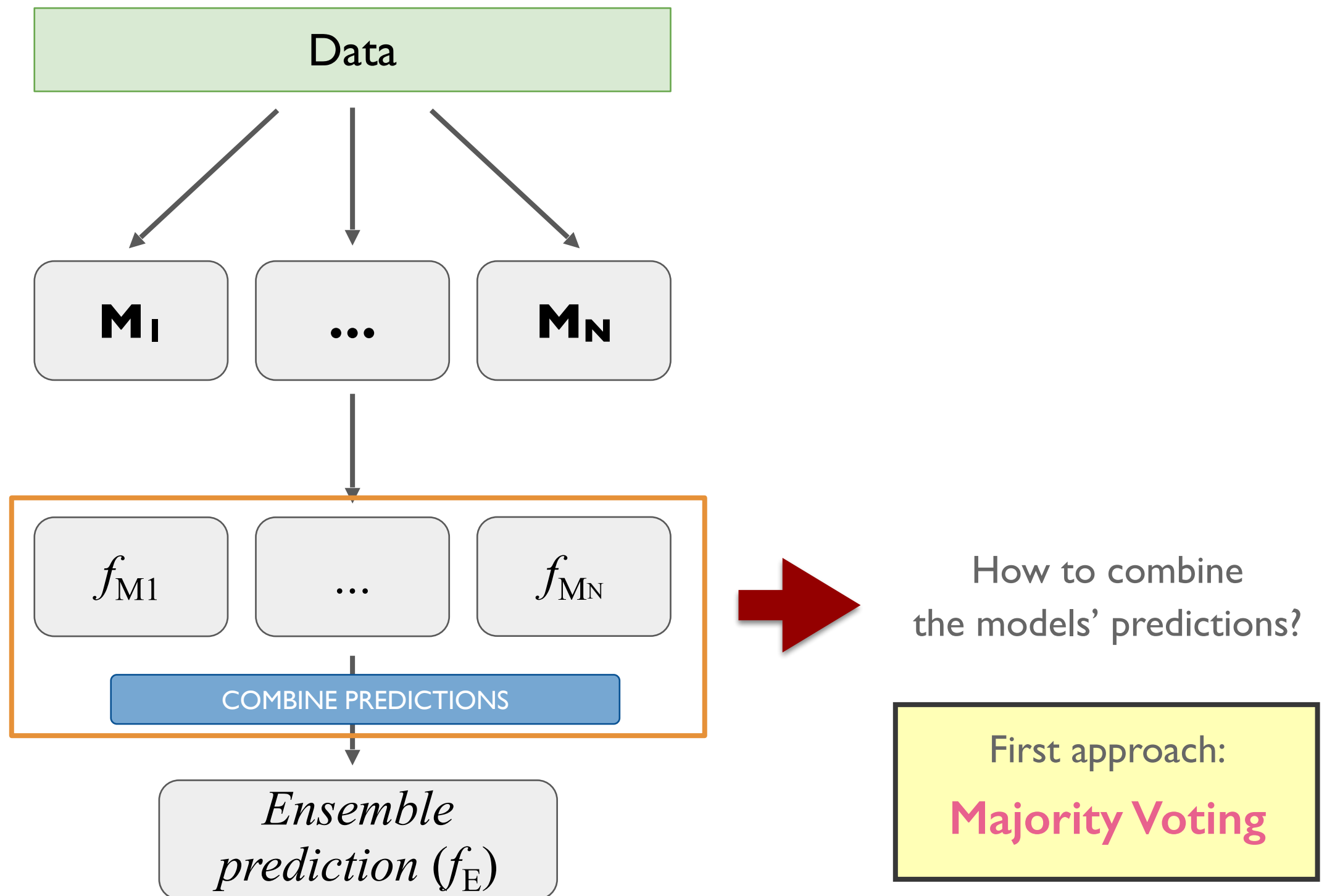
“The ensemble consensus tends to have higher accuracy than that of its individual classifiers”

Conditions for *ensembles* to perform well:  
“**accuracy and diversity**”

- **Error rate** of individual classifiers  $< 50\%$
- **Errors made by classifiers are independent**
- Under these conditions:
  - if classifiers have similar error rate (e.g., 45%)
  - the expected error rate of the ensemble decreases linearly with the number of models

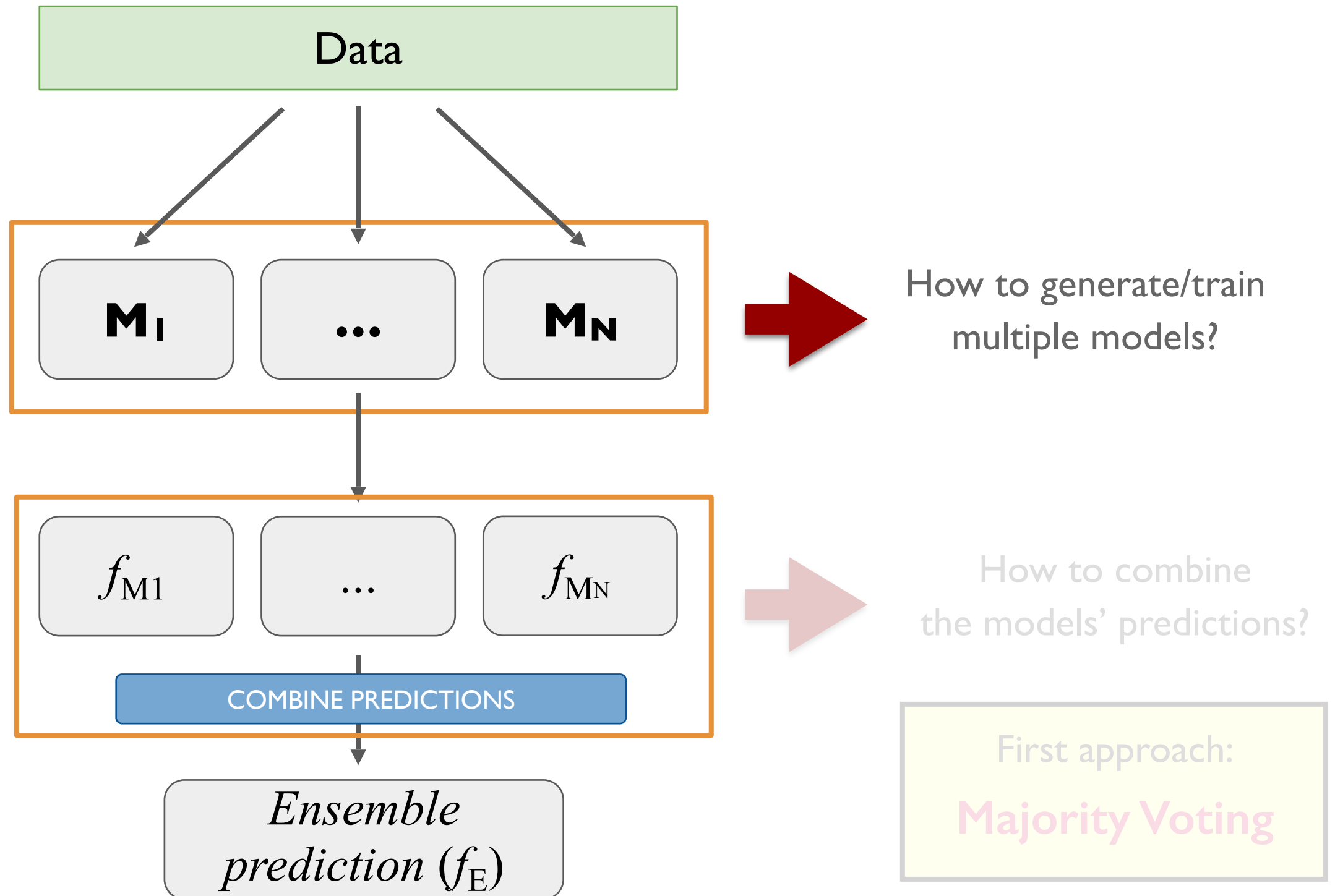


# Multiple Models in ML





# Multiple Models in ML





# Multiple Models in ML

Data

**Ideally: generate a *diverse* set of models in the ensemble**

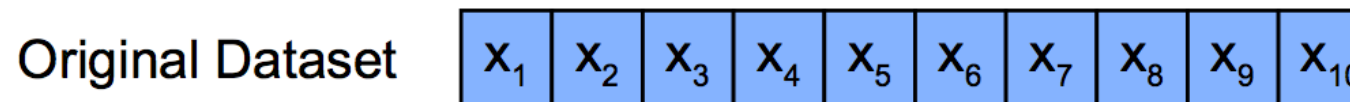
**What does diversity mean?**

**Methods based on (re-)sampling from the training set**

*Ensemble  
prediction ( $f_E$ )*

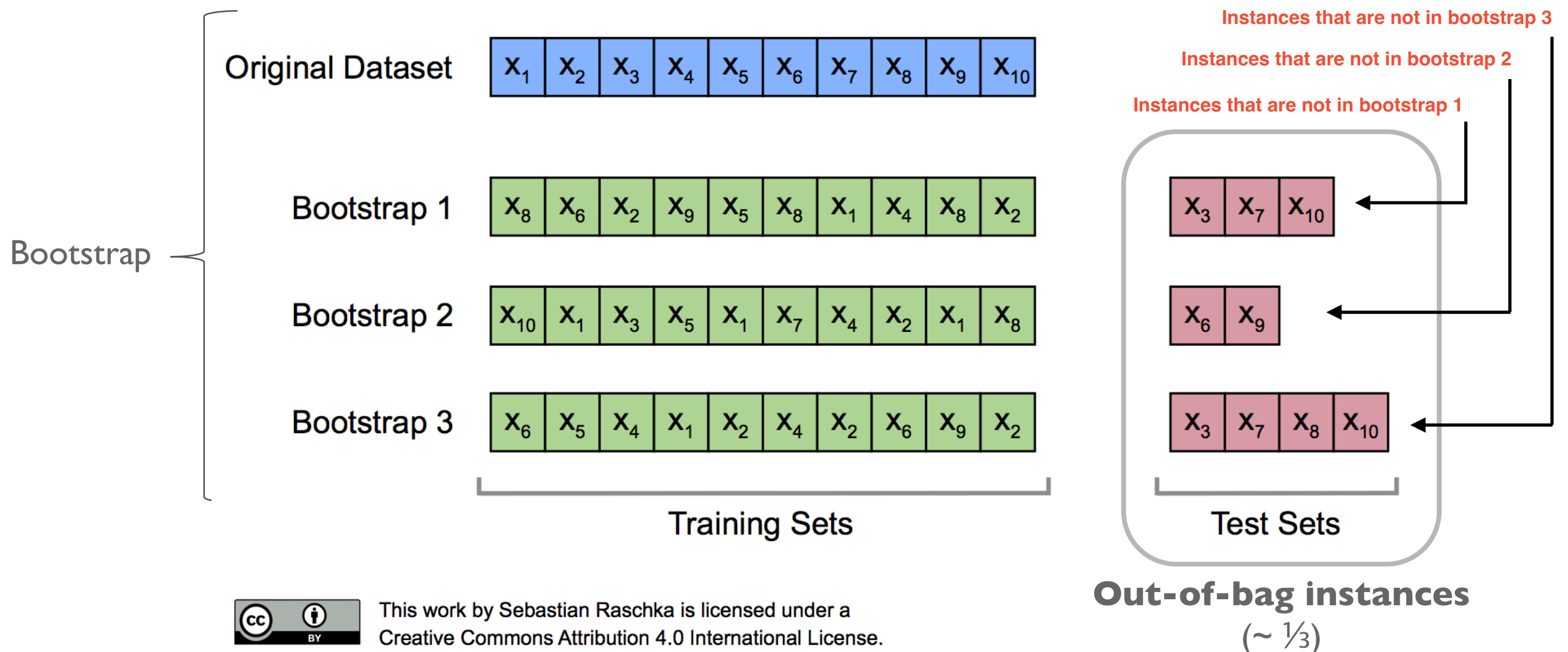
# Bagging (a.k.a., Bootstrap Aggregating)

- Diversity: achieved by generating many “synthetic” datasets based on the original training set
  - Each new “synthetic” dataset is known as a **bootstrap** dataset
  - Bootstrap datasets are created by **sampling (with replacement)** from the original/complete dataset
  - If the original dataset has size  $N$ , **each bootstrap dataset should also have size  $N$**



# Bagging (a.k.a., Bootstrap Aggregating)

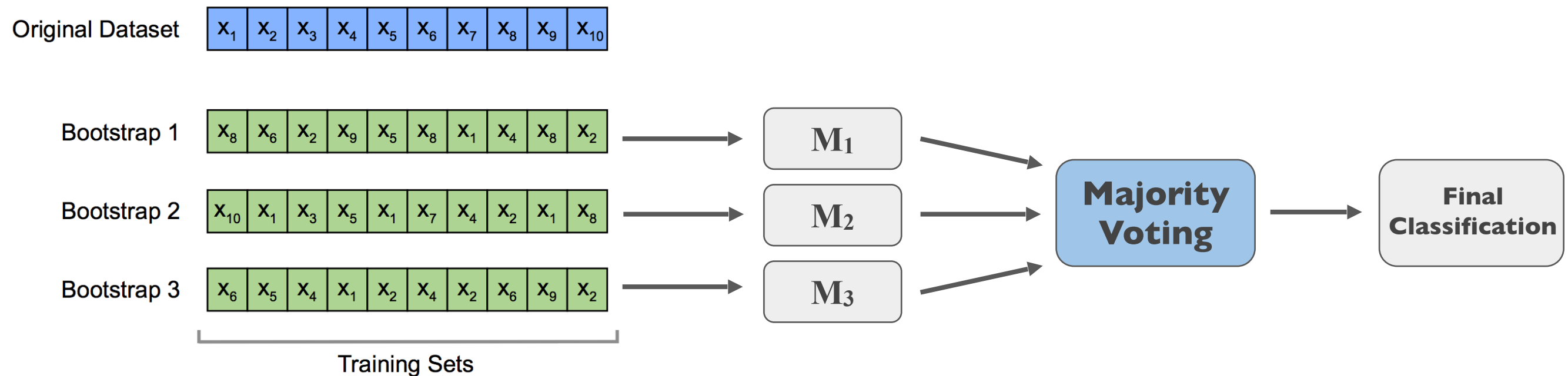
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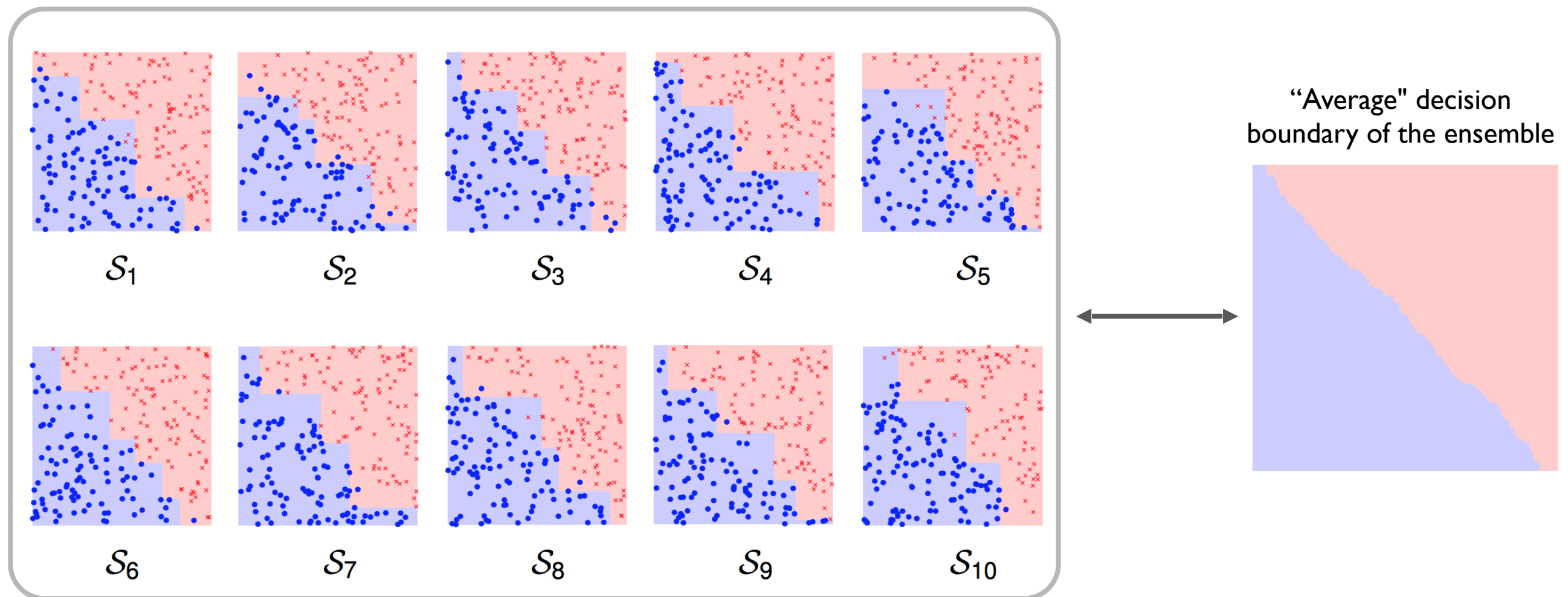
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  - Bootstrap datasets are created by **sampling (with replacement)** from the original/complete dataset
  - If the original dataset has size  $N$ , each bootstrap dataset should also have size  $N$
  - One classifier (same algorithm, same hyper-parameters) is trained based on each bootstrap
  - To classify new instances, perform majority voting considering all models in the ensemble



# Bagging (a.k.a., Bootstrap Aggregating)

- How would Bagging work if combined with the Decision Tree algorithm?
  - Train multiple decision trees
  - Each one based on a different bootstrap dataset (e.g., bootstrap datasets  $\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_{10}$ )
  - Because data given to each tree is slightly different...
    - the trees will be slightly different (e.g., will perform different tests/attribute splits)
  - Each tree will thus result in a slightly different decision boundary





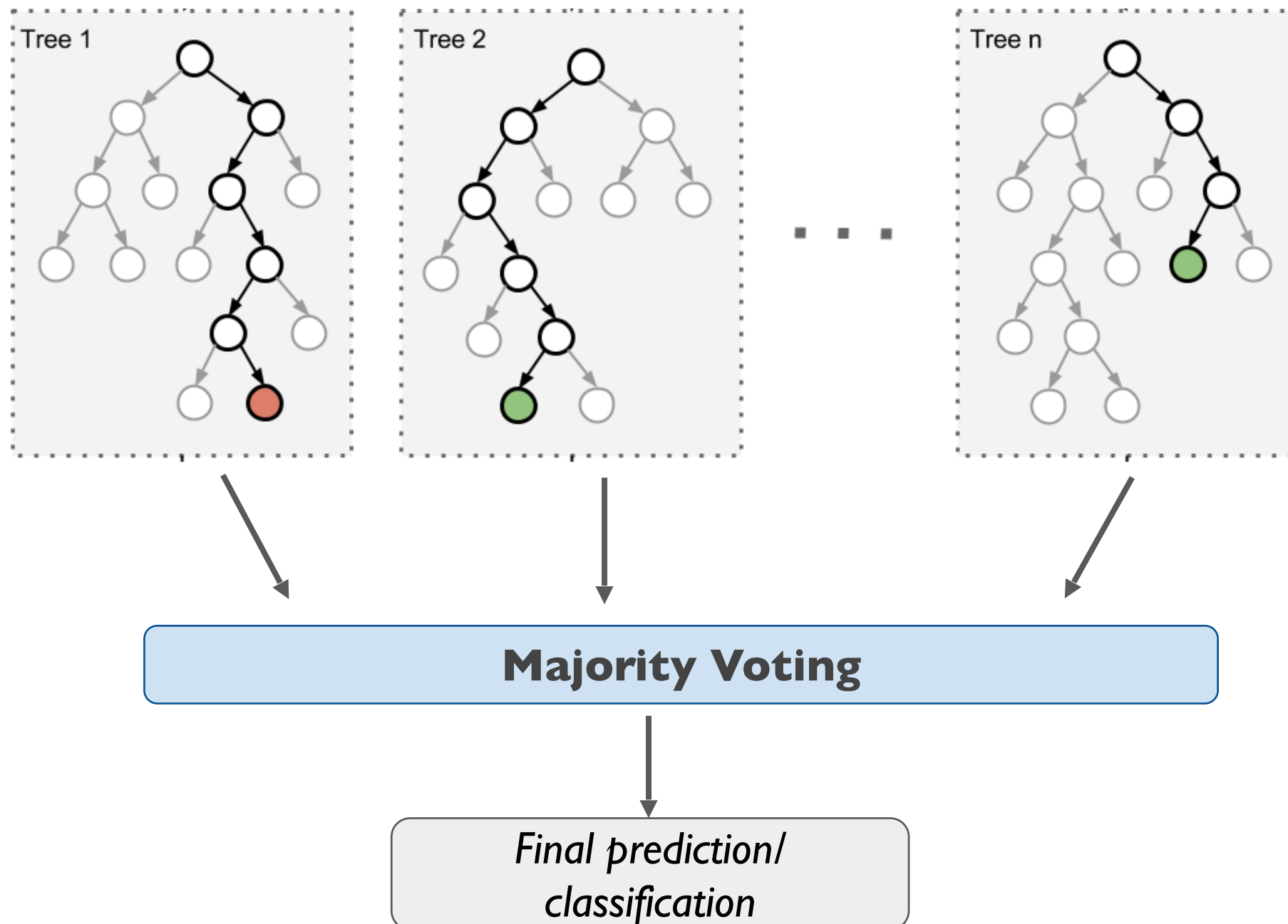
# Random Forests





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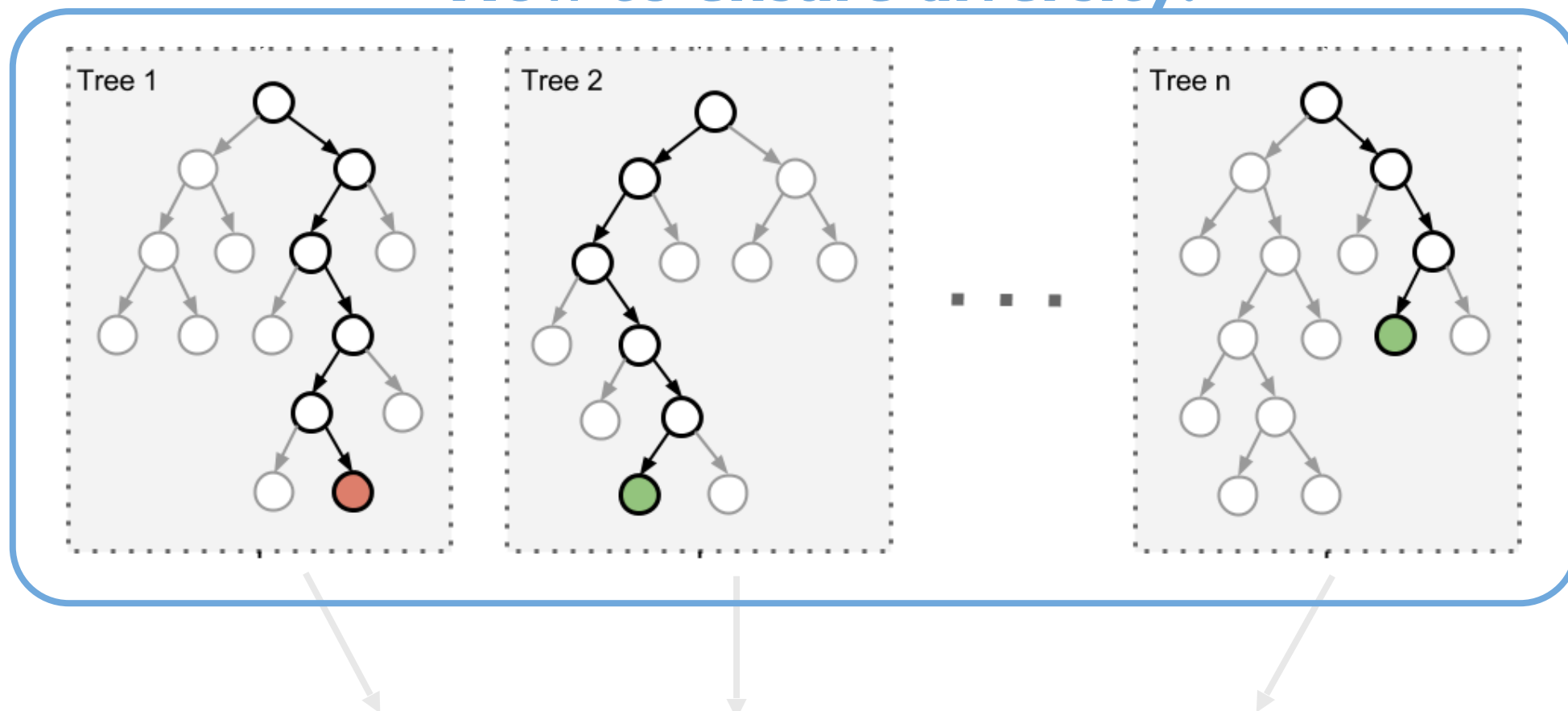
- Trains many (slightly different) Decision Trees
- Their predictions are combined via majority voting



# Random Forests

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## How to ensure diversity?



**Bagging, combined with selection of random attributes**

*Final prediction/  
classification*



# Random Forests

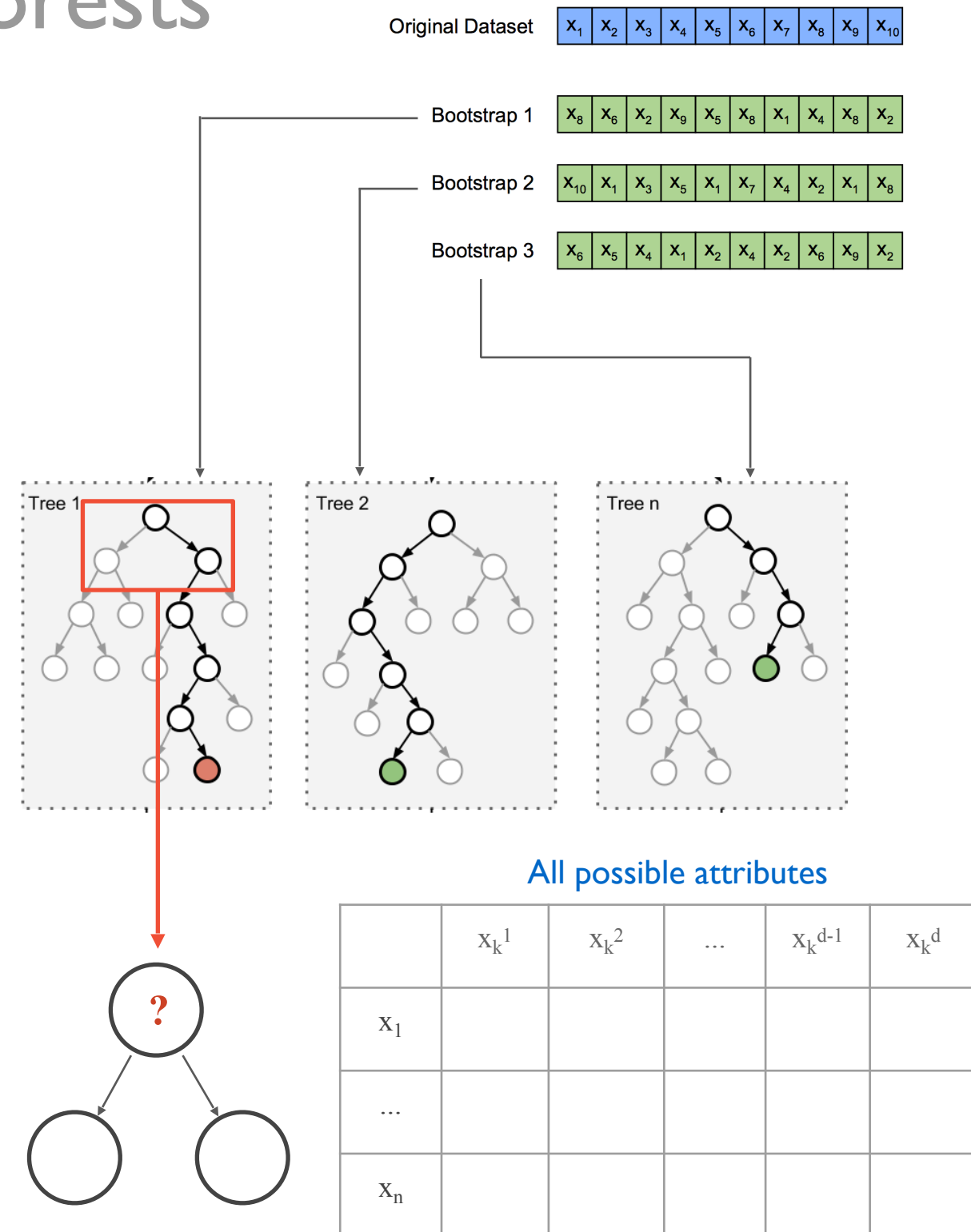
- **Bagging + selection of random attributes**

- **Bagging**

- Each tree is trained with a different bootstrap dataset
  - Sampling (with replacement) from original dataset

- **Selection of random attributes**

- Whenever splitting a node
  - Does not check Information Gain of *all* attributes
  - Only of a subset of *m* randomly selected attributes,
    - Out of all possible attributes



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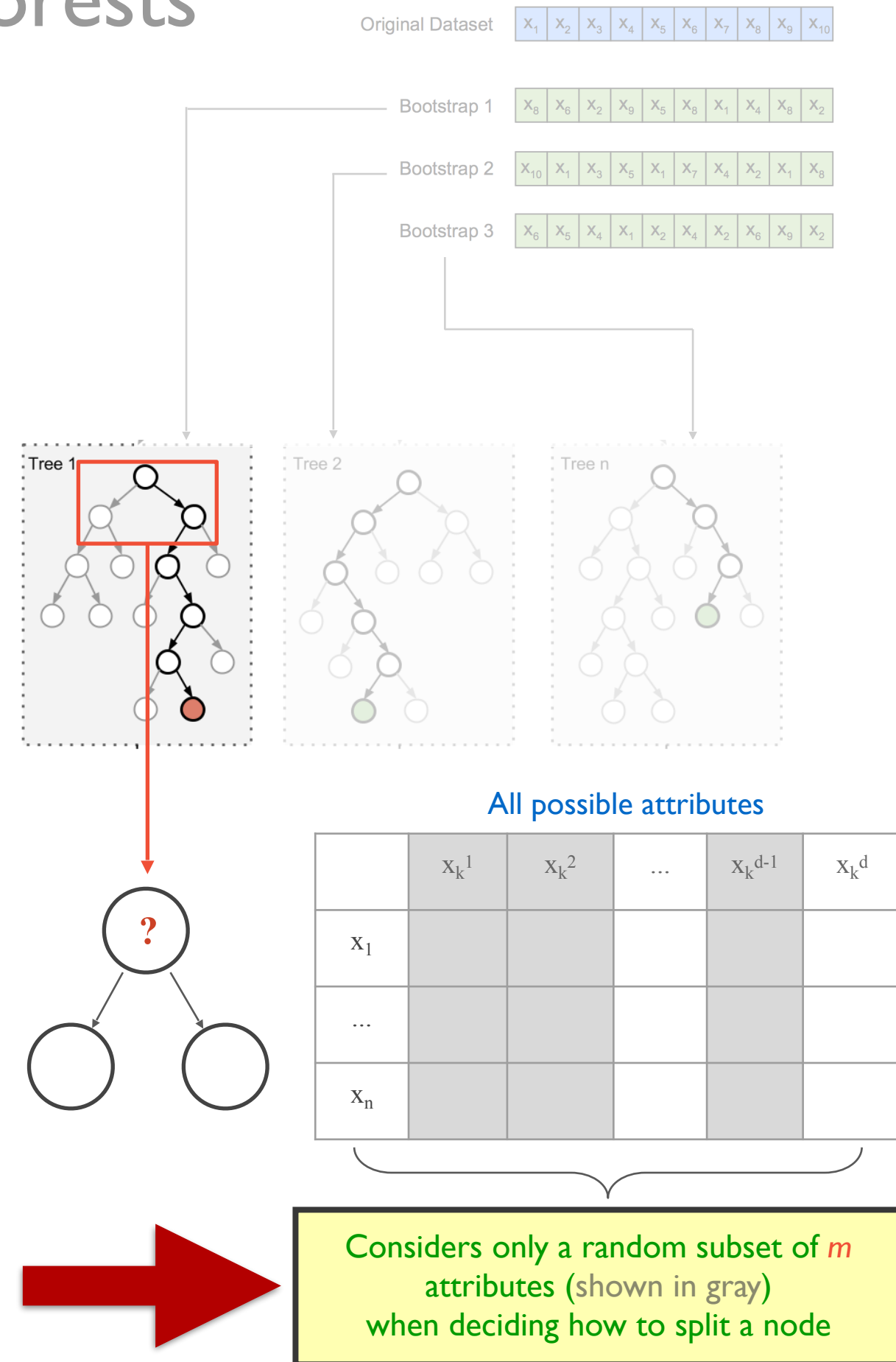
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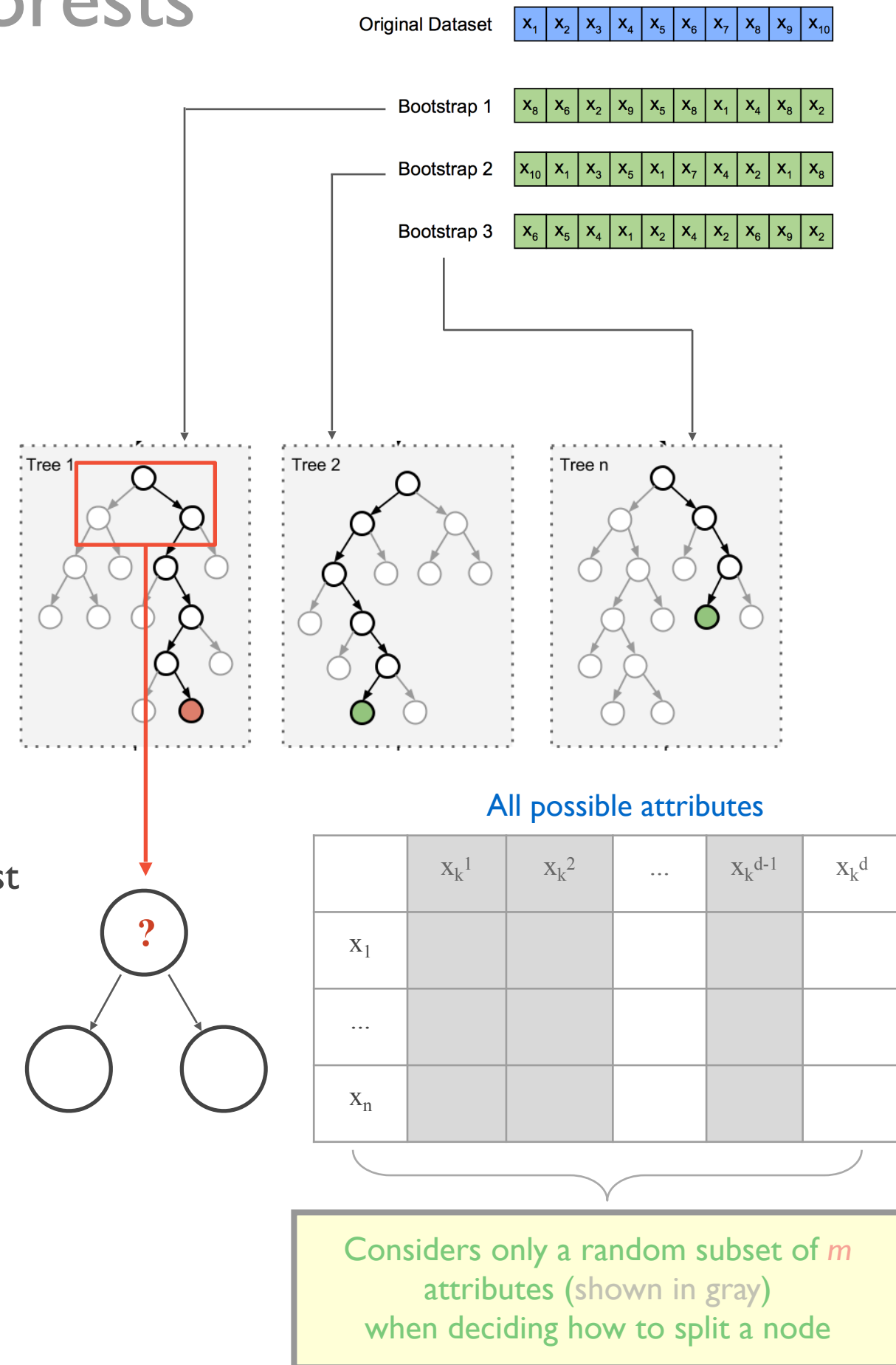
- Whenever splitting a node
  - Does not check Information Gain of *all* attributes
  - Only of a subset of *m* randomly selected attributes,
    - Out of all possible attributes
    - Splits node based on the attribute with best Information Gain (or Gini index, etc)

- **Performing these two sampling procedures**

- Highly diverse ensemble of decision trees

- **Out-of-Bag instances (test instances)**

- Approximately 1/3 of the data
- Used to evaluate the ensemble

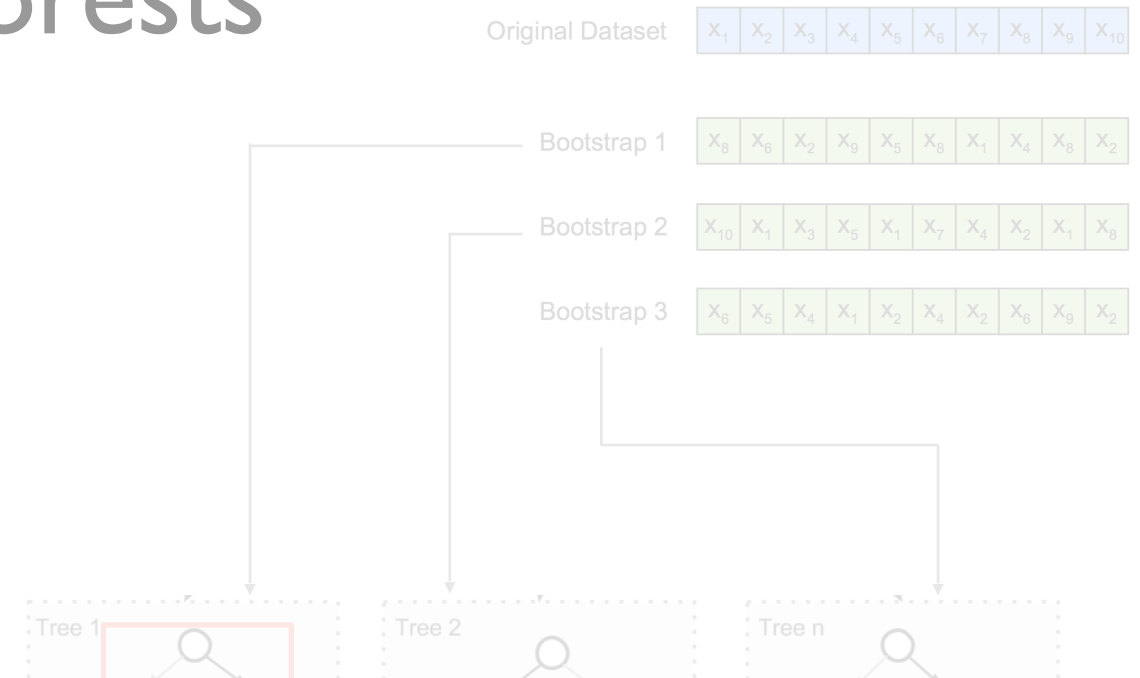


# Random Forests

- **Bagging + selection of random attributes**

- **Bagging**

- Each tree is trained with a different bootstrap dataset
  - Sampling (with replacement) from original dataset



One of the most widely used (supervised) machine learning model nowadays

Often, state-of-the-art in real-life applications

- Out of all possible attributes
- Splits node based on the attribute with best Information Gain (or Gini index, etc)

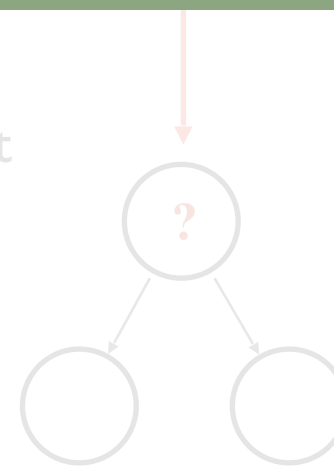
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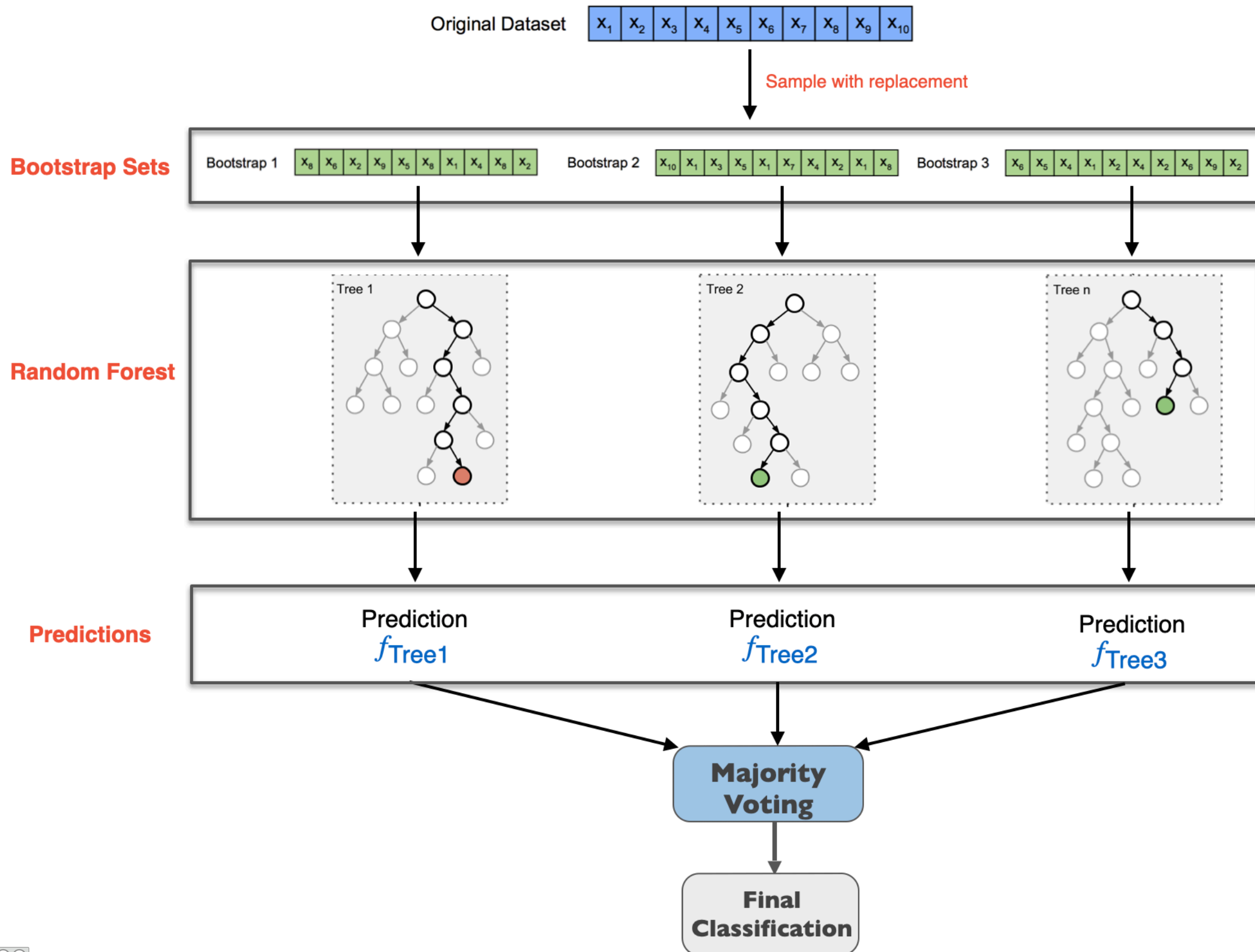


All possible attributes

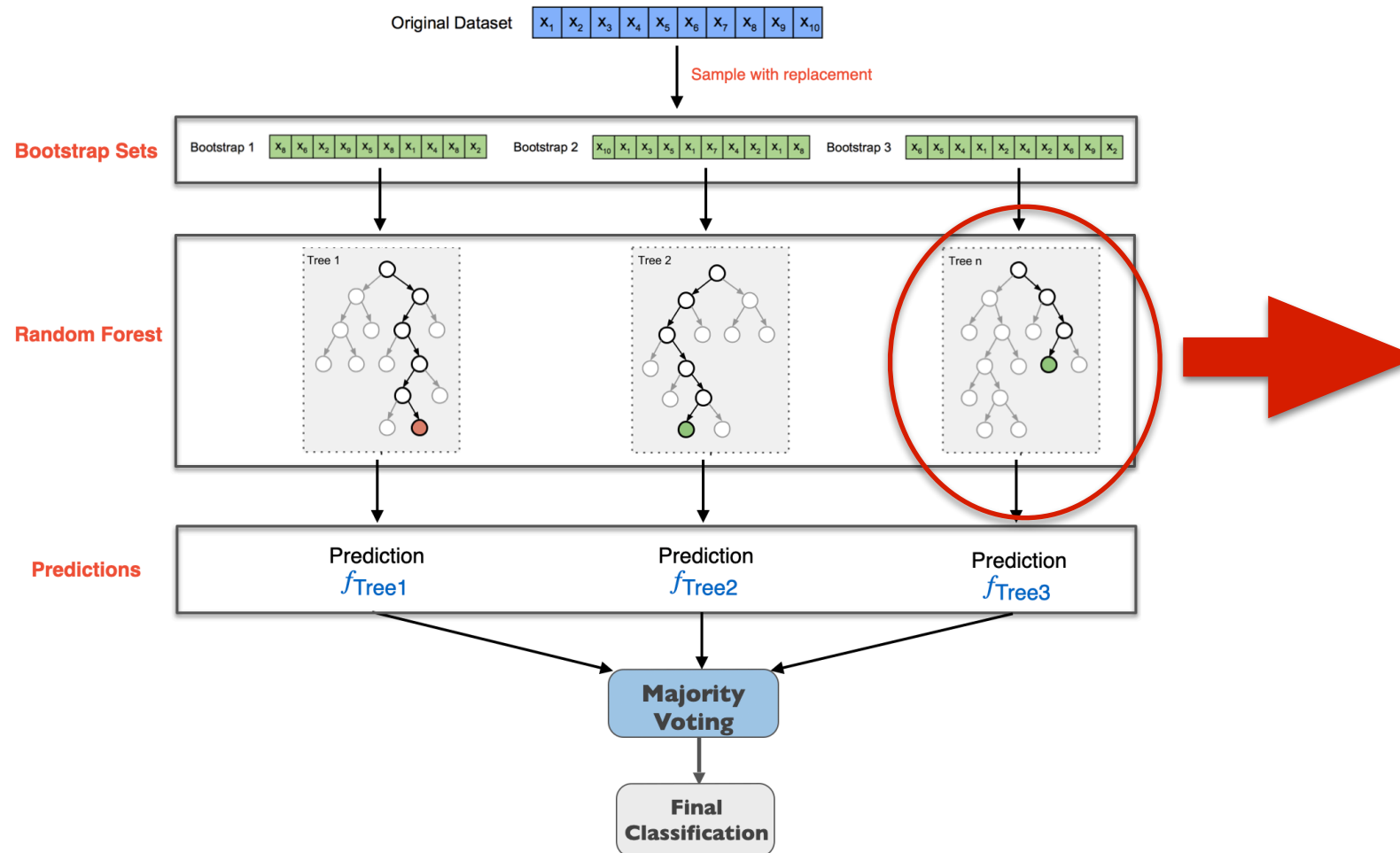
|       | $x_k^1$ | $x_k^2$ | ... | $x_k^{d-1}$ | $x_k^d$ |
|-------|---------|---------|-----|-------------|---------|
| $x_1$ |         |         |     |             |         |
| ...   |         |         |     |             |         |
| $x_n$ |         |         |     |             |         |

Considers only a random subset of  $m$  attributes (shown in gray) when deciding how to split a node

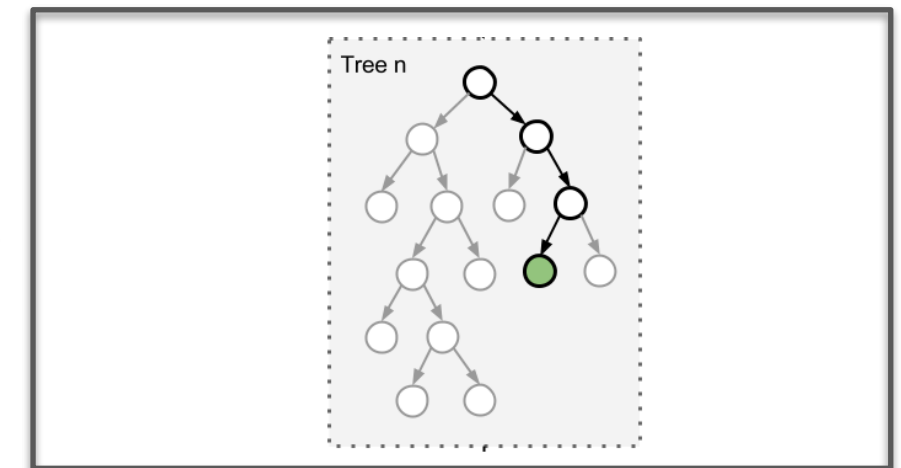
# Random Forests



# Random Forests



## A Random Tree



1. Assume we are splitting the **green node**
2. Assume there are a total of  $X$  features
3. Pick a random subset of  $m \approx \sqrt{X}$  of all attributes
4. Out of these, select the one with best Information Gain



# Random Forests: Training

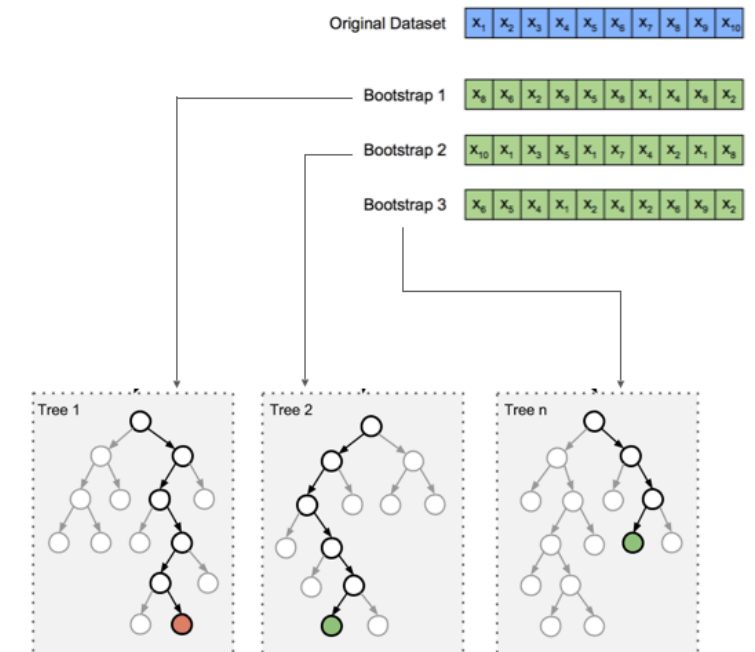
## 1. Let $D$ be the original training set

1.  $D$  contains  $N$  training instances, each with  $X$  attributes

## 2. For each bootstrap $b = 1, \dots, B$

1. Construct a bootstrap dataset of size  $N$  by sampling from  $D$  with replacement
2. Train a decision tree based on this bootstrap by recursively:
  1. Picking a random subset of  $m \approx \sqrt{X}$  attributes
  2. Out of these, select the best attribute to split the current node  
(e.g., based to Information Gain)
  3. Add the node to the tree and use it to partition the data into disjoint subsets

## 3. Return the ensemble of learned trees $\rightarrow$ the Random Forest



# Machine Learning

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### Bagging Random Forests

